SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 5

Strategic Capacity Planning

Strategic capacity planning (Ch. 5) plus strategy lenses (RBV/VRIN and dynamic capabilities) and the Pelindo & Semarang Container Terminal case. Covered: capacity as a strategic commitment; capacity, best operating level and utilisation; economies and diseconomies of scale; focus and flexibility; step size and timing; determining requirements (forecast → convert → project → cushion → chunk → balance); decision trees (Hackers, undiscounted and at 16%); service capacity and utilisation zones; RBV & VRIN; Five Forces → RBV → dynamic capabilities; and six TPKS discussion questions.

Material

1 Learning objectives and roadmap

1. Explain capacity management and why the level is a strategic commitment.
2. Plan capacity: requirements, cushions, chunk size, system balance.
3. Evaluate alternatives with decision trees.
4. Compare service and manufacturing capacity through RBV and dynamic capabilities.

2 Why capacity is a strategic problem

- Capacity is the output a system can achieve over a period — a rate.
- Strategic capacity planning sets capital-intensive resources for long-term strategy.
- Horizons: long (>1 yr), intermediate (6–18 months), short (<1 month).
- Measure in outputs if the mix is stable, inputs if not.

Lead time
Bought against a forecast.

Irreversibility
Shedding means selling or liquidating.

Asymmetry
Too little loses customers; too much forces discounting.

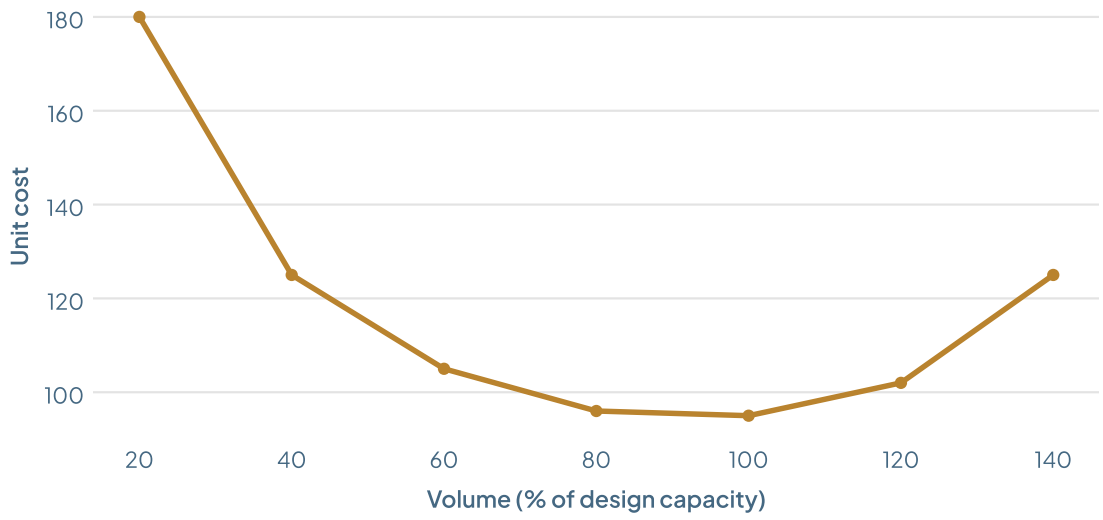
3 Capacity, best operating level and utilisation

Term	Definition
Capacity	Achievable output over a period — always state the period.
Best operating level	Volume minimising average unit cost.
Utilisation rate	Capacity used ÷ best operating level.

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Utilisation = used ÷ best operating level → 480 ÷ 500 = 96%
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Below: fixed cost spread thin; at: lowest cost but no cushion; above: overtime, wear, defects — and queues in services.

Average unit cost vs volume (conceptual)



4 Economies and diseconomies of scale

1. **Economies of scale** — lower cost per unit as volume rises.
2. **Turning point** — keeping a huge facility busy may need discounting; an M&M line moves 2.6 million sweets an hour with heavy maintenance labour.
3. **Diseconomies** — coordination and complexity push cost up again.

Plant size
Often set by transport cost (a cement plant serves a local market).

! **Scale ≠ productivity**
Jaguar ~15 cars per employee; Volvo Torslanda 29; BMW Mini 39 with 4,500 workers.

5 Capacity focus and capacity flexibility

A **focused factory** pursues a limited set of objectives. Ways: plant within a plant; flexible plants (near-zero changeover); flexible processes (economies of scope); flexible workers.

PWP zone	Process	Focus
High volume, low variety	Flow line	Cost
Low volume, high variety	Job shop	Flexibility
New product ramp	Pilot cell	Learning
Service & spares	Fast response	Delivery

Economies of scope; several best levels under one roof; flexibility is bought up front.

6 Determining requirements, cushions and chunks

1. Forecast
2. Convert to equipment and labour
3. Project against availability
4. Cushion
5. Chunk
6. Balance

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Capacity cushion = (capacity - demand) ÷ demand
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Example

Demand $10m, capacity $12m → 20% cushion = 83% utilisation; demand $12m vs capacity $10m → –16.7%.

Trade-off	Frequent small	Infrequent large
Idle capacity cost	Low	High
Shortage risk	Higher	Lower
Scale economies	Forgone	Captured
Disruption	Repeated	One-off
Capital	Staged	Front-loaded

7 Considerations in changing capacity

Consideration	Explanation
System balance	Add at the bottleneck, buffer in front of it, or duplicate.

Frequency	Too often = repeated disruption; too rarely = large chunks as overhead.
External sources	Outsourcing; sharing (airlines swap aircraft).
Decreasing	Harder than adding; permanent cuts mean selling or liquidating.

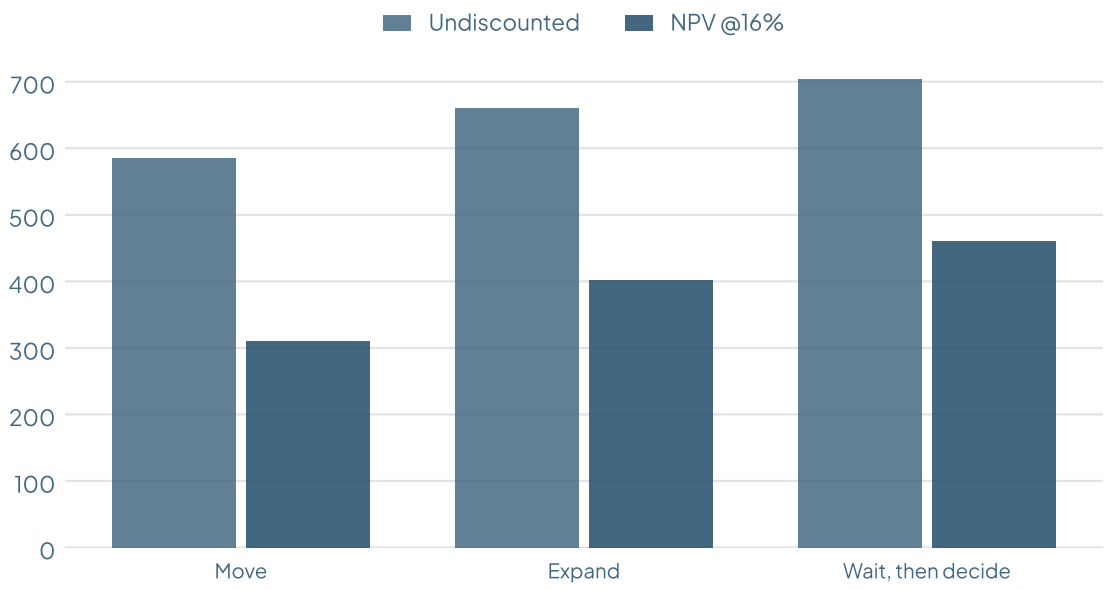
8 Decision trees: Hackers Computer Store

Squares are decisions, circles are chance events. Hackers (5 years, 55% strong growth):

Option	Cost	Per year (strong / weak)	EV
Move	$210,000	$195,000 / $115,000	$585,000
Expand	$87,000	$190,000 / $100,000	$660,500
Do nothing (option to expand in year 2)	$0	$170,000 / $105,000	$703,750

Solve right to left; the tree also says don't expand in year two. At 16%: $310,613, $402,507, $460,857 (–47%, –39%, –35%) — ranking holds, margins compress, timing matters.

Hackers: expected values ($000)

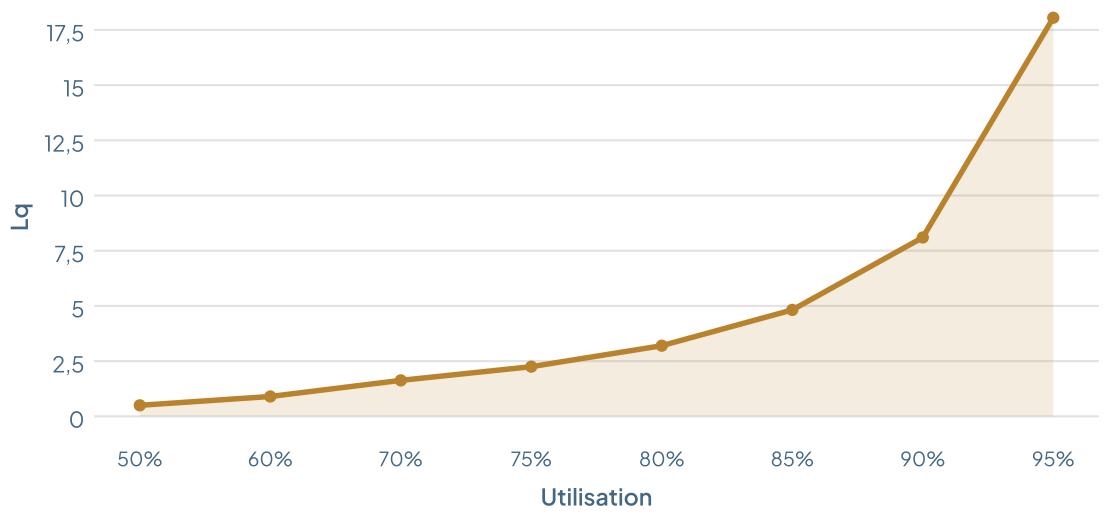


9 Service capacity is a different problem

Dimension	Manufacturing	Services
Storability	Inventory buffers	Capacity perishes hourly
Location	Produce centrally, distribute	Distribute capacity first
Volatility	Smoothed	Higher
Planning increment	Weekly/monthly	10–30 minutes
Utilisation target	Near 100%	~70%
Running full	Unit cost rises	Queue and quality fall first
Lever	Build, subcontract	Reprice peaks, reshape demand, extend hours

A service at 47% annual utilisation can still turn customers away at 9 a.m. Zones: idle, service (best ~70%), critical, non-service.

Queue length vs utilisation (single server)



Low targets for ERs; ~70% for most services; near 100% for commuter trains; full for stadiums.

10 The resource-based view and the VRIN test

Test	Question	If no
Valuable	Exploit opportunity / neutralise threat?	Disadvantage
Rare	Few competitors hold it?	Parity
Inimitable	Costly to copy?	Temporary advantage
Non-substitutable	No equivalent route?	Temporary advantage
Organised	Structured to capture value?	Unexploited potential

Sequential test. Capacity anyone can buy is valuable but not rare → parity; resources that pass usually constrain capacity (site, right-of-way, licence, relationship).

11 From Five Forces to dynamic capabilities: sensing, seizing, transforming

Framework	Advantage	Capacity question
Five Forces (outside-in)	Industry structure	Will the industry pay for the capacity?
RBV (inside-out)	VRIN assets	Is it rare and hard to copy?
Dynamic capabilities	Reconfiguring the asset base	Can we reconfigure if wrong?

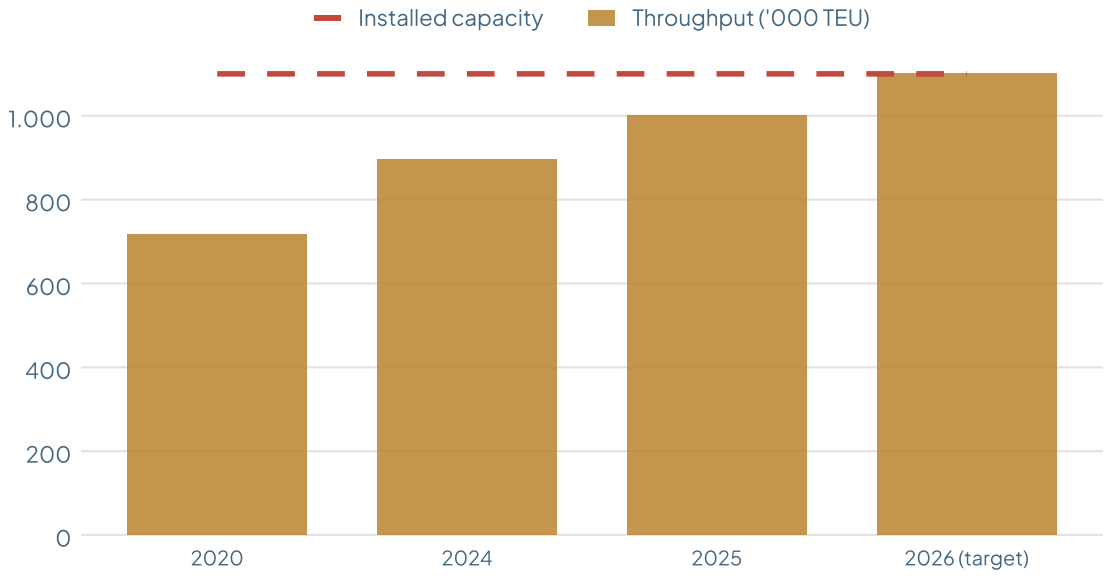
Teece: ordinary capabilities (do things right, imitable); **sensing** (scan, interpret), **seizing** (business model, alliances, investment protocols), **transforming** (restructure, redeploy, divest) — a continuous cycle.

Capacity as orchestration: never neutral; sunk-cost rule; binding constraint may be land, licence or relationship; conversion often beats construction. Moves: sense before you size, seize in defensible chunks, transform before you duplicate.

12 Pelindo & Semarang Container Terminal: adding capacity on sinking ground

September 2026: Andira Prawitasari has three weeks before the 2027–2031 capital proposal. TPKS ~1m TEU (2025), installed 1.1m TEU = 2026 target; ~70% of Central Java boxes ship via Priok and Perak.

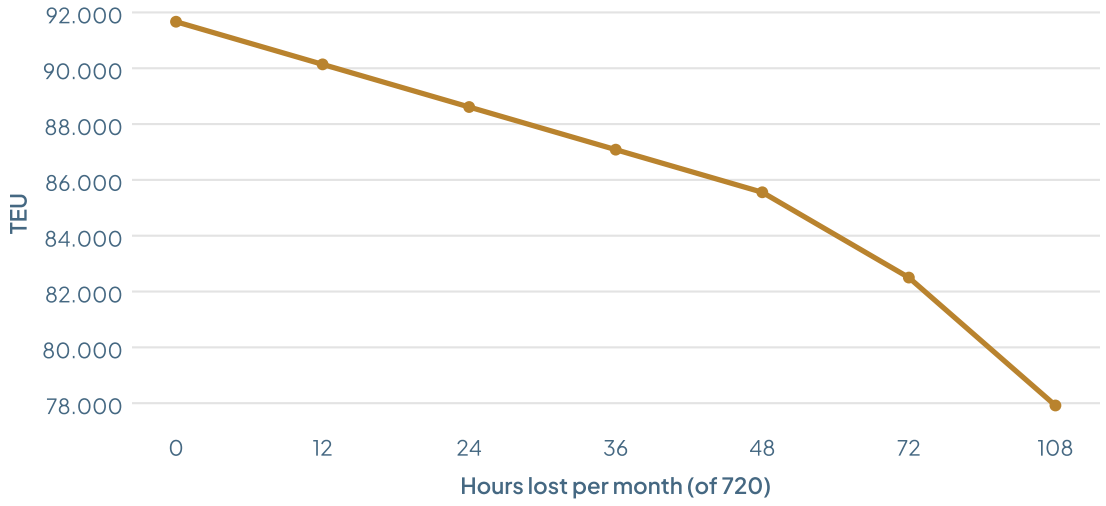
TPKS throughput ('000 TEU) and utilisation



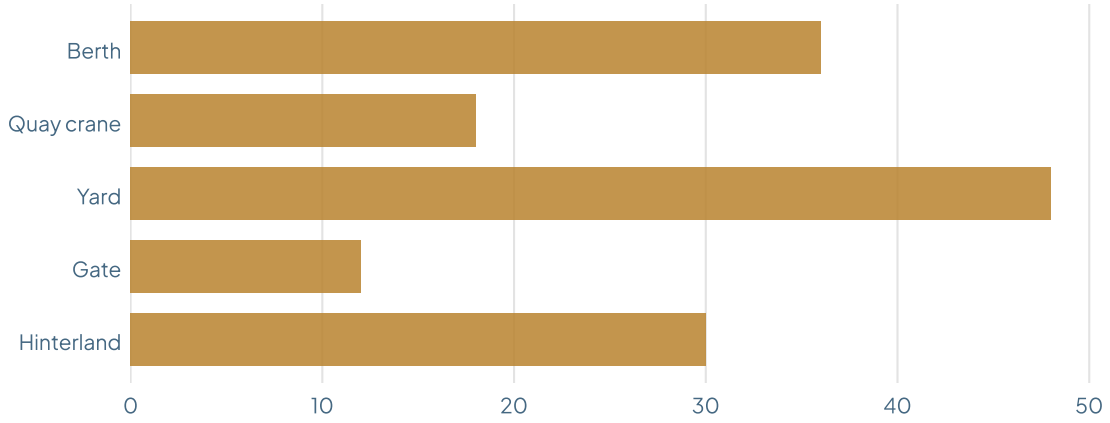
Utilisation 65.2% → 81.4% → ~90.9% → 100%.

Demand: 717,062 TEU (2020) → 895,904 (2024) → >1m (2025); 1.7m circulating for 2027 vs an official 1.2m by 2029. Ground: subsidence of a dozen-plus cm a year; May 2025 sea-wall collapse (~25 m), ~75% of the port flooded, berth halted. Demand moving west: Kendal 100% taken up, KITB Batang ~60%; two ~50 ha dry-port candidates.

Effective monthly capacity vs flood hours lost



Investment lead time by stage (months, midpoint)



Midpoints of the case ranges.

Binding question

Is TPKS really the binding stage, or is it ground elevation and hinterland choice?

Discussion: state the period; add at the bottleneck (yard, outside-the-fence constraints); bring subsidence in as net available operating hours; decision tree with irreversible branches behind triggers; VRIN — the permitted site is rare precisely because it can't move; design a routine with a trigger number, small reversible commitments and open options.

13 Debrief: six things from the Pelindo case

1. State the period out loud.
2. Cushion is a decision.
3. Find the bottleneck before sizing the asset.
4. The binding constraint may not be machine-shaped.
5. Chunk and timing are the real decision — on sinking ground, too large costs more than too small.
6. The source of advantage can be the limit.

Key takeaways

1. Capacity is a rate over a stated period; three horizons.
2. Lead time, irreversibility, asymmetry.
3. Best operating level and utilisation (480 ÷ 500 = 96%).
4. Economies → diseconomies; transport sets plant size; scale ≠ productivity.
5. Focused factory, PWP, flexibility, economies of scope.
6. Forecast → convert → project → cushion → chunk → balance.
7. Balance, frequency, external sources, decreasing.
8. Decision trees: Hackers EVs and 16% NPVs.
9. Service capacity and the ~70% target.
10. VRIN(O).
11. Five Forces → RBV → dynamic capabilities.
12. TPKS lessons.

Case studies

Case & worked answers

Pelindo & Semarang Container Terminal: capacity on sinking ground

Throughput grew to >1m TEU on 1.1m capacity, giving a zero 2026 cushion; subsidence >12 cm/yr and a May 2025 flood halted operations. Q1: annual utilisation hides peaks — measure weekly YOR, dwell time, vessel waits. Q2: the yard is the least elastic stage; cranes alone move the queue ashore; the hinterland may bind. Q3: effective capacity = installed × net available operating hours index (720 – 36)/720 = 95% → ~87,083 TEU/month. Q4: full B is irreversible and risks stranding; the hybrid has the best EV/downside. Q5: the Tanjung Emas site is VRIN yet caps capacity. Your syndicate's recommendation:

sequential hybrid with Chunk 1 (US$8–12m), Chunk 2 dry-port JV (US$20–25m), YOR ≥85% trigger and subsidence ≥15 cm or downtime ≥10% kill criteria. Improvement: quantify probabilities and EVs.

Worked examples

Problem 1 · Easy · **Utilisasi dan cushion**

Lini produksi: best operating level 100 unit/jam, output aktual 85 unit/jam. Pabrik lain: kapasitas 1.200 unit/hari, permintaan 1.000 unit/hari. Hitung utilisasi lini pertama dan capacity cushion pabrik kedua.

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Utilisasi = Kapasitas terpakai ÷ Best operating level = 85 ÷ 100 = 85%
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Cushion = (Kapasitas - Permintaan) ÷ Permintaan = 200 ÷ 1.000 = 20%
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Answer: Utilisasi 85%; cushion 20%.

Problem 2 Medium **Kapasitas efektif saat banjir**

Kapasitas terpasang 100.000 TEU/bulan pada 720 jam terjadwal. Bulan ini 54 jam hilang karena banjir rob. Berapa kapasitas efektif?

1. Index = (720 − 54) ÷ 720 = 0,925.
2. Kapasitas efektif = 100.000 × 0,925 = 92.500 TEU.

Answer: 92.500 TEU (hilang 7.500 TEU).

Problem 3 Medium **Decision tree ekspansi**

Permintaan tinggi p = 0,6, rendah 0,4. Ekspansi besar: 500 / −100. Ekspansi kecil: 300 / 100 (Rp juta). Mana yang dipilih?

1. EV besar = 0,6 × 500 + 0,4 × (−100) = 300 − 40 = 260.
2. EV kecil = 0,6 × 300 + 0,4 × 100 = 180 + 40 = 220.
3. Besar unggul 40, tetapi berisiko rugi 100 jika permintaan rendah.

Answer: Ekspansi besar (EV 260 vs 220) — kecuali perusahaan tidak sanggup menanggung kemungkinan rugi Rp100 juta.

Practice questions

Try answering first. The answer key is below each question.

1. Capacity 1.1m TEU, throughput 1.0m. Utilisation…

A. 81%
B. 90.9%
C. 100%
D. 110%

Answer: B. 1.0/1.1.

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2. Adding cranes when the yard is full…

A. Lifts throughput
B. Moves the queue ashore
C. Cuts dwell time
D. Reduces floods

Answer: B. Add at the bottleneck.

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3. Face-to-face services typically target about…

A. 40%
B. 70%
C. 95%
D. 100%

Answer: B. Above that, queues grow.

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4. A resource anyone can buy (cranes) gives, per VRIN…

A. Sustained advantage
B. Parity
C. Temporary advantage
D. Disadvantage

Answer: B. Valuable but not rare.

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5. 'Transforming' is best illustrated by…

A. Building a duplicate plant
B. Converting an underused gas line to oil
C. Overtime
D. Raising prices

Answer: B. Energy East.

Sources

- P3MD MK4 OSCM — Strategic Capacity Planning (class slides)
- Jacobs & Chase — OSCM Ch. 5; Barney (1991); Teece (1997, 2007); Pelindo & TPKS case